Class-2

Question and answers:

1. Find the Fourier Cosine transform of $f(t) = \sinh(at)$, $a > 0$. Also find the inverse finite Fourier Cosine transform.

Ans: $$C_0 = \int_0^{\pi} \sinh(at)dt = \frac{\cosh(at)}{a}\Bigg]_0^{\pi} = \frac{\cosh a\pi - 1}{a}$$

$$C_n = \int_0^{\pi} \sinh(at)\cos(\text{n}\,t)dt$$

$$= \frac{1}{2}\int_0^{\pi} (e^{at} - e^{-at})\cos nt\, dt$$

$$= \frac{1}{2}\left[\left\{\frac{e^{at}}{a^2+n^2}(a\cos nt + n\sin nt)\right\}_0^{\pi} - \left\{\frac{e^{-at}}{a^2+n^2}(-a\cos nt + n\sin nt)\right\}_0^{\pi}\right]$$

$$= \frac{1}{2(a^2+n^2)}\left[e^{a\pi} a\cos n\pi - a + e^{-a\pi} a\cos n\pi - a\right]$$

$$C_n = \frac{a}{2(a^2+n^2)}\left[2\,\text{cosha}\,\pi\cos n\pi - 2\right] = \frac{a}{(a^2+n^2)}\left[\text{cosha}\,\pi\cos n\pi - 1\right].$$

Inverse finite Fourier Cosine transform is $f(t) = \frac{1}{\pi}C_0 + \frac{2}{\pi}\sum_{n=1}^{\infty} C_n\cos(nt)$

$$f(t) = \frac{1}{\pi}\left(\frac{\cosh a\pi - 1}{a}\right) + \frac{2}{\pi}\sum_{n=1}^{\infty}\left(\frac{a}{(a^2+n^2)}\left[\text{cosha}\,\pi\cos n\pi - 1\right]\right)\cos(nt)$$

2. Find the Fourier sine transform of $f(t) = e^{-t}$, if the function is defined on $[0, \pi]$.

Ans: $$S_n = \int_0^{\pi} f(t)\sin(nt)dt$$

$$= \int_0^{\pi} e^{-t}\sin(nt)\,dt$$

$$= \left[\frac{e^{-t}}{1+n^2}(-\sin(nt) - n\cos(nt))\right]_0^{\pi}$$

$$= \frac{1}{1+n^2}\left[e^{-\pi}(-n)\cos(\text{n}\,\pi) + \text{n}\right]$$

$$= \frac{n}{1+n^2}\left[1 - e^{-\pi}\cos(n\pi)\right]$$